Residual-Aware Neural Spectral Capacity for Transformer Architecture Selection
CS5491 Artificial Intelligence – Project Proposal

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Problem and motivation. We choose Direction 1, Stronger Transformer-block Modeling. Neural Spectral Capacity (NSC) cheaply ranks architectures before training by summing matrix capacities, but does not explicitly model the addition of a branch output to its input. We will study one mechanism: the identity shortcut in an FFN residual sub-block. Our question is whether modeling this addition improves trained-architecture ranking and selection under a parameter budget. Improvement is a hypothesis, not a promised outcome.

Method and assumptions. Our reference model linearizes the residual sub-block as $J = I + B$, where $B = W2\ D\ W1$ is the FFN branch Jacobian. We assume independent zero-mean Gaussian projection weights with standard deviation 0.02 and independent Bernoulli(0.5) diagonal activation gates D; normalization is approximated by a fixed identity mapping. These are simplifying reference assumptions, not claims about the archive's exact architecture. For hidden width d, define $C(A) = \ln \det(I + A^T A)$. We propose the expected residual contribution $C(I + B)$ minus $d \ln 2$, subtracting the identity-only capacity so a zero branch contributes zero. We will determine a specification-only estimator through development-stage numerical checks. The ablation uses expected $C(B)$, with the same branch model but no shortcut; original NSC remains a separate baseline. Attention scoring, initialization and projection grouping stay fixed. Checks will cover zero and weak branches, small sampled matrices, and estimator error and cost. Layer contributions remain additive: reversing their order must leave the score unchanged, so this study does not claim order sensitivity.

Data and evaluation. We will use the supplied 200 archived GPT-2/LiteTransformerSearch architectures and fixed 100-development/100-final split, inspecting configurations and freezing all choices before final evaluation. Baselines comprise original NSC, the shortcut ablation, the positional heuristic, reported parameter count and reference FLOPs. We will report Spearman correlation, Kendall tau-b against negative perplexity, ties and scoring time, plus comparisons at equal hidden width and within 10% total parameters, including their counts. At parameter caps of 10M, 20M, 30M and 50M, each method selects $k = 1$ and 3 from identical feasible pools. We will report best selected perplexity and regret against the feasible optimum, alongside repeated uniform-random selection. Heterogeneous configurations limit causal interpretation; one outcome per architecture cannot establish training-seed uncertainty.

Feasibility and schedule. The archive study and numerical checks form the complete core project. Before planning new training, we will time a small decoder run and record hardware, throughput and peak memory. Only if affordable will we add matched-architecture training with fixed data splits, parameter budgets and training settings. The starter has not yet been obtained, and no experiments or training have been run. By November 19 we will deliver reproduced baselines, the estimator and development results; by December 3, frozen final evaluation and reproducible code. No LLM is used experimentally: model and access are not applicable; expected LLM API calls: 0.